

7. Acquisition C: disk and cutout area series

Series C measures sensed area against true area for disks and cutouts. Each plate is captured at many random sub-pixel offsets and at every station of the ladder. Sensed area depends on where a feature's edge falls relative to the pixel grid and the projector dots, and on the subtended size D_{px} , which the Z sweep varies by a factor of 4 for each feature. The procedure averages over the phase and also measures its spread.

1. The pose list for C holds, for each plate, T4 (disks) and T5 (cutouts), and each gap, 15 and 60 mm (GAP_SMALL_MM, GAP_LARGE_MM), every station of the ladder, with the plate centered and fronto-parallel. Mount each plate with the mount check of Step 4.8; the plan tool orders the poses so that plates are re-mounted as rarely as possible and randomizes the order within each mounting.
2. At each station, capture 30 poses (PHASE_JITTER_POSES_AREA). Each has a logged random lateral offset, uniform over ± 4 px, half the 8 px jitter span (PHASE_JITTER_SPAN_PX), in H and V, with 10 frames (FRAMES_PER_AREA_POSE).
3. Field sub-series. At the reference station, 800 mm (Z_REFERENCE_MM), with the small gap, repeat Step 2 for each plate at the four off-axis field positions with 10 poses each (FIELD_SUBSERIES_POSES_AREA). The plates do not fit at the full field offset there, so the plan tool pulls them inward as Section 5, Step 1 describes and logs the fraction achieved; the C field comparison reports it.
4. Open-background variant (cutouts, optional). At the reference station, remove the back plate so that nothing lies within the sensor's range behind the holes: at least $Z_{MAX} - Z_{REFERENCE_MM}$, 800 mm, of clear space behind the plate, or a surface beyond the sensor's range. Then repeat Step 2. This separates the sensor's fill-in behavior from reads of a real back surface.
5. Capture drift sentinels on the mounted plate's front surface, as Section 5, Step 3 defines them, every 60 minutes (DRIFT_SENTINEL_INTERVAL_MIN).